MATH 223: Linear Algebra

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Abstract

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1 Introduction

2 Prerequisite Knowledge

2.1 Introduction to Proof

2.1.1 Direct Proof

The basic idea of a direct proof of $P \implies Q$ is:

1. Assume P is true,
2. Use P to show that Q is true (the hard part).

Example. *An integer number n is odd if and only if there exists an integer k such that $n = 2k + 1$.*

Proof. Assume a, b are consecutive integers, meaning $b = a + 1$. So

$$a + b = a + (a + 1) = 2a + 1.$$

Thus b is odd if and only if a is odd. □

2.1.2 Proof by Contradiction

The idea is that a proposition is either true or false, but not both. We get a contradiction when we can show a statement is both true and false, showing our initial assumptions are inconsistent. We use this to show $P \implies Q$ by assuming both P and not Q are simultaneously true and deriving a contradiction.

Example. *An integer number n is odd if and only if there exists an integer k such that $n = 2k + 1$.*

Proof. Assume a and b are consecutive integers. Assume that $(a + b)$ is not odd. If $(a + b)$ is not odd, then no integer k such that $(a + b) = 2k + 1$. But $a + b = a + (a + 1) = 2a + 1$. We have shown that $(a + b) \neq 2k + 1$ but $(a + b) = 2a + 1$, a contradiction. □

2.1.3 Proof by Contrapositive

The idea is to prove $P \implies Q$ by instead proving not $Q \implies$ not P.

Example. *If n^2 is even, then n is even.*

Proof. We show the contrapositive: if n is odd, then n^2 is odd. Assume n is odd, $\exists k_1 \in \mathbb{Z}, n = 2k_1 + 1$. So

$$n^2 = (2k_1 + 1)^2 = 4k_1^2 + 4k_1 + 1 = 2(2k_1^2 + 2k_1) + 1.$$

Let $k_2 = 2k_1^2 + 2k_1$. Then $n^2 = 2k_2 + 1$, so n^2 is odd. □

2.2 Notation

2.2.1 Sets

Sets are a grouping of objects.

Set	Meaning	Examples
$\mathbb{N}$	The set of natural numbers	(0, 1, 2, 3, ...)
$\mathbb{Z}$	The set of integers	(..., -3, -2, -1, 0, 1, 2, 3, ...)
$\mathbb{Q}$	The set of rational numbers	$\mathbb{Q} = \frac{a}{b} \mid \forall a, b \in \mathbb{Z}\ and\ b \neq 0$
$\mathbb{R}$	The set of all rational and all irrational numbers	$(..., -1, 0, \frac{1}{4}, 1, 1000, ...)$
$\mathbb{C}$	The set of all complex numbers	$\mathbb{C} = \{x + iy \mid x, y \in \mathbb{R}\ and\ i \subseteq \sqrt{-1}\}$.

We have the following relationships between sets:

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}.$$

2.2.2 Symbols

Symbol	Meaning
$\subseteq$	is a subset of or equal to
$\subset$	is a strict subset of
$\in$	is an element of
$\forall$	for all
$\exists$	there exists
$\emptyset$	empty set
$\Rightarrow$	implies
$\Leftrightarrow$	if and only if
$\cong$	is isomorphic to[1]

2.3 Complex Numbers

2.3.1 Definition and the Fundamental Theorem of Algebra

Definition 1 (Complex Number)**.** *A complex number is of the form: $z = x + iy$ where $x, y \in \mathbb{R}$ and i is the imaginary unit $i = \sqrt{-1}$.*

Theorem 1 (Fundamental Theorem of Algebra)**.** *The FTA states that any non-constant, single-variable polynomial[2] with complex coefficients has at least one root in $\mathbb{C}$.*

Remark 1. *If we have a polynomial f of degree n, then it has n roots, where each root can have a multiplicity[3].*

[1] is isomorphic to = structurally the same as.

[2] A polynomial is a function such as $f = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ where $a_i \in \mathbb{R}$ or $\mathbb{C}$ and $n \in \mathbb{N}$.

[3] The multiplicity of a root represents how many times the root occurs in the polynomial.

Example. $(x-1)^2$ *has degree 2 but only one root, which is 1, with multiplicity 2.*

We can factorize a polynomial $f = a_n x^n + \cdots + a_1 x + a_0$ as $f = a(x-z_1)(x-z_2)\cdots(x-z_n)$, where z_i are the roots of f in $\mathbb{C}$.

2.3.2 Arithmetic Operations

We define the following operations on complex numbers $z = x + iy$ and $z' = x' + iy'$:

- Addition: $z + z' = (x + x') + i(y + y')$.
- Multiplication: $zz' = (xx' - yy') + i(xy' + yx')$.
- Inverse: $\dfrac{1}{z} = \dfrac{\overline{z}}{z\overline{z}} = \dfrac{x - iy}{x^2 + y^2} = \dfrac{x}{x^2 + y^2} + i\dfrac{-y}{x^2 + y^2}$.

Multiplying by a complex number z corresponds geometrically to:

$$\begin{cases} \text{a rotation by angle } \theta, \\ \text{a rescaling by factor } |z|. \end{cases}$$

2.3.3 Complex Conjugate

Definition 2 (Complex Conjugate)**.** *For $z = x + iy$, the complex conjugate is $\overline{z} = x - iy$.*

Basic properties:

- $\overline{\overline{z}} = z$
- $\overline{z + z'} = \overline{z} + \overline{z'}$
- $\overline{z \cdot z'} = \overline{z} \cdot \overline{z'}$

2.3.4 Geometric Interpretation and Modulus

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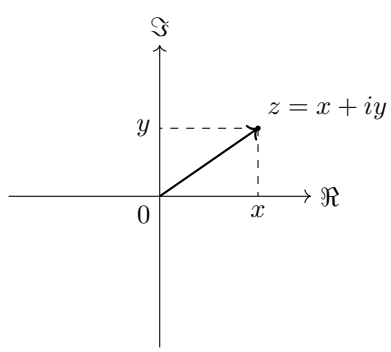
Definition 3 (Geometric interpretation)**.** *Every complex number $z = x + iy$ is identified with a point (x, y) in the complex plane:*

$$\mathbb{C} = \{x + iy \mid x, y \in \mathbb{R}\}.$$



Definition 4 (Modulus)**.** *The modulus of z is*

$$|z| = \sqrt{z\bar{z}} = \sqrt{x^2 + y^2}.$$

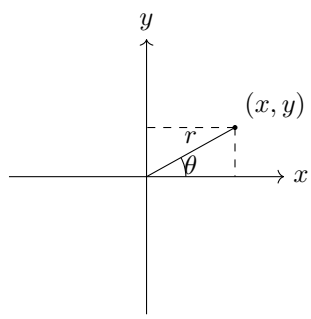
Geometrically, $|z|$ is the distance from the origin to (x, y).

The unit circle in the complex plane can be written as:

$$S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\} = \{z \in \mathbb{C} : |z| = 1\}.$$

2.3.5 Polar Coordinates and Polar Form

Definition 5 (Polar coordinates)**.** *Instead of (x, y), a point can be described by polar coordinates (r, θ), where r is the distance to the origin and θ is the angle with the positive x-axis.*



So $x = r\cos\theta$ and $y = r\sin\theta$, with $r = |z|$.

Definition 6 (Euler's formula)**.**

$$e^{i\theta} = \cos\theta + i\sin\theta. \tag{1}$$

Definition 7 (Polar and exponential form)**.** *If $z = x + iy$ with $r = \sqrt{x^2 + y^2}$, then $x = r\cos\theta$, $y = r\sin\theta$, so*

$$z = r(\cos\theta + i\sin\theta) = re^{i\theta}.$$

Definition 8 (Multiplication in polar form)**.**

$$z = re^{i\theta}, \quad z' = r'e^{i\theta'} \implies zz' = rr'e^{i(\theta+\theta')}.$$

Example.

$$(1 + i)^{32} = (\sqrt{2}\, e^{i\pi/4})^{32} = 2^{16}\, e^{i8\pi} = 2^{16}(\cos 8\pi + i\sin 8\pi) = 2^{16}.$$

2.3.6 Roots in $\mathbb{C}$

Definition 9 (n^{th} roots)**.** *For $z \in \mathbb{C}$, an nth root of z is $w \in \mathbb{C}$ with $w^n = z$.*

Example. *If $z = re^{i\theta}$, the solutions of $w^n = z$ are*

$$w_k = r^{1/n} e^{i(\theta+2\pi k)/n}, \quad k = 0, 1, \ldots, n-1.$$

They are equally spaced on a circle of radius $r^{1/n}$ centered at the origin.

Definition 10 (Roots of unity)**.** *The nth roots of unity are solutions of $w^n = 1$:*

$$w_k = e^{i2\pi k/n}, \quad k = 0, 1, \ldots, n-1.$$

They lie on the unit circle and are equally spaced.

2.3.7 Hermiticity

Definition 11 (Hermitian Matrix)**.** *A matrix $A \in M_n(\mathbb{C})$ is Hermitian if $A = A^*$, where $A^* = \overline{A}^T$. Equivalently,*

$$a_{ij} = \overline{a_{ji}} \quad \forall i, j.$$

Example. $A = \begin{pmatrix} 2 & 1+i \\ 1-i & 3 \end{pmatrix}$ *is Hermitian since $A^* = A$.*

Remark 2. *All eigenvalues of a Hermitian matrix are real, and eigenvectors corresponding to distinct eigenvalues are orthogonal.*

Problem 1 (Hermitian Matrix)**.** *Let $A, B \in M_n(\mathbb{C})$. Prove that $(AB)^* = B^* A^*$.*

Definition 12 (Anti-Hermitian Matrix)**.** *A matrix $A \in M_n(\mathbb{C})$ is anti-Hermitian if $A = -A^*$, where $A^* = \overline{A}^T$. Equivalently,*

$$a_{ij} = -\overline{a_{ji}} \quad \forall i, j.$$

Problem 2 (Hermiticity and Anti-Hermiticity)**.** *Prove that A is Hermitian iff iA is anti-Hermitian.*

2.4 Basic Algebraic Structures

2.4.1 Sets with Multiplication

Definition 13 (Set with multiplication)**.** *A set M is a set with multiplication if for any $a, b \in M$, the product $ab \in M$.*

Example. *$M_2(\mathbb{C})$ (all 2×2 complex matrices) and $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$ are sets with multiplication.*

2.4.2 Invertibility

Definition 14 (Invertibility)**.** *There are two methods of defining invertibility:*

- *Let $A \in M_n(k)$. If there exists $B \in M_n(k)$ with $AB = I_n$ or $BA = I_n$, then A is invertible and $B = A^{-1}$. Here I_n is the $n \times n$ identity matrix*[4]*.*
- *Let $A \in M_n(k)$ with $\det(A) \neq 0$. Then the inverse of A is*

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A),$$

where $\operatorname{adj}(A)$ is the adjugate of A.

Definition 15 (Adjugate of a Matrix)**.** *The adjugate of $A \in M_n(k)$ is the transpose of the cofactor matrix of A:*

$$\operatorname{adj}(A) = \operatorname{cof}(A)^t,$$

where cof *is the cofactor matrix. For a 2×2 matrix this reduces to:*

$$\operatorname{adj}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Definition 16 (Cofactor Matrix)**.** *The cofactor matrix of $A \in M_n(k)$ is the matrix C whose (i, j) entry is the cofactor*

$$C_{ij} = (-1)^{i+j} M_{ij},$$

where M_{ij} is the determinant of the matrix obtained by deleting row i and column j from A. The sign pattern $(-1)^{i+j}$ is:

$$\begin{pmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

Example. *Let* $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{pmatrix}$. *The cofactors are:*

$$C_{11} = (+1)\det\begin{pmatrix} 4 & 5 \\ 0 & 6 \end{pmatrix} = 24, \quad C_{12} = (-1)\det\begin{pmatrix} 0 & 5 \\ 1 & 6 \end{pmatrix} = 5,$$

$$C_{13} = (+1)\det\begin{pmatrix} 0 & 4 \\ 1 & 0 \end{pmatrix} = -4, \quad C_{21} = (-1)\det\begin{pmatrix} 2 & 3 \\ 0 & 6 \end{pmatrix} = -12,$$

$$C_{22} = (+1)\det\begin{pmatrix} 1 & 3 \\ 1 & 6 \end{pmatrix} = 3, \quad C_{23} = (-1)\det\begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix} = 2,$$

$$C_{31} = (+1)\det\begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix} = -2, \quad C_{32} = (-1)\det\begin{pmatrix} 1 & 3 \\ 0 & 5 \end{pmatrix} = -5,$$

$$C_{33} = (+1)\det\begin{pmatrix} 1 & 2 \\ 0 & 4 \end{pmatrix} = 4.$$

[4] The identity matrix has 1s on its diagonal and 0s elsewhere.

The cofactor matrix is

$$\operatorname{cof}(A) = \begin{pmatrix} 24 & 5 & -4 \\ -12 & 3 & 2 \\ -2 & -5 & 4 \end{pmatrix}$$

Transposing the cofactor matrix to obtain the adjugate:

$$\operatorname{adj}(A) = \begin{pmatrix} 24 & -12 & -2 \\ 5 & 3 & -5 \\ -4 & 2 & 4 \end{pmatrix}.$$

One can verify: $\det(A) = 22$*, and indeed* $A^{-1} = \frac{1}{22}\operatorname{adj}(A)$.

Remark 3. *If A is invertible, A^{-1} is unique*[5]*.*

Example. $2 \in \mathbb{Z}$ *is* not *invertible since* $\frac{1}{2} \notin \mathbb{Z}$*, but* $2 \in \mathbb{R}$ *is invertible since* $\frac{1}{2} \in \mathbb{R}$.

Example. $1 + i \in \mathbb{C}$ *is invertible:* $\dfrac{1}{1+i} = \dfrac{1-i}{2} \in \mathbb{C}$.

Problem 3 (Invertibility)**.** *Show that if an inverse of A in M exists, then it is unique.*

Problem 4 (Invertibility 2)**.** *Let K be a field. Prove that* $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \in M_2(K)$ *is not invertible.*

2.4.3 Rings

Definition 17 (Ring)**.** *A ring is a set R with addition and multiplication satisfying:*

1. $(R, +)$[6] *is a commutative*[7] *and associative*[8] *group with zero ($\exists 0 \in R$) and additive inverses (every element has an additive inverse $-a$ with $a + (-a) = 0$). Essentially, R behaves like $\mathbb{Z}$ under addition.*

2. *Multiplication is associative with identity* 1*. This means $(ab)c = a(bc)$ for all $a, b, c \in R$, and there exists $1 \in R$ with $1 \cdot a = a \cdot 1 = a$. Note that multiplication need* not *be commutative (i.e., ab may not equal ba), and nonzero elements need not have multiplicative inverses.*

3. *Distributivity: $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$.*

Example. $(\mathbb{Z}, +, \cdot)$ *is a ring: it is closed under addition and multiplication, has zero element and additive inverses, multiplication is associative with identity 1, and multiplication distributes over addition.*

[5] Unique: there is exactly one such element.

[6] The notation $(R, +)$ means the set R equipped with the operation $+$. It specifies which operation we are talking about, since R has both $+$ and $\times$ defined on it.

[7] $a + b = b + a$.

[8] $a + (b + c) = (a + b) + c$.

2.4.4 Fields

Definition 18 (Field)**.** *A field is a ring* (17) *in which every nonzero element has a multiplicative inverse.*

Example. $(\mathbb{Q}, +, \cdot)$ *is a field: it is a ring, and every nonzero* $q \in \mathbb{Q}$ *has* $q^{-1} \in \mathbb{Q}$ *with* $q \cdot q^{-1} = 1$.

Remark 4. $\mathbb{Z}$ *is a ring but not a field: for example, there is no integer* x *with* $2x = 1$.

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3 Vector Spaces

Definition 19 (Vector Space)**.** *Let* k *be a field. A vector space over* k *is a set* V *with addition and scalar multiplication satisfying:*

1. *Closure under addition:* $v_1 + v_2 \in V$
2. *Commutativity:* $v_1 + v_2 = v_2 + v_1$
3. *Associativity:* $(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3)$
4. *Zero vector:* $\exists\, 0 \in V$ *with* $v + 0 = v$
5. *Additive inverses:* $\forall v \in V, \exists\, -v \in V$ *with* $v + (-v) = 0$
6. *Closure under scalar multiplication:* $\lambda v \in V$
7. *Multiplication associativity:* $\lambda(\mu v) = (\lambda\mu)v$
8. *Identity:* $1 \cdot v = v$
9. *Distributivity over vectors:* $\lambda(v_1 + v_2) = \lambda v_1 + \lambda v_2$
10. *Distributivity over scalars:* $(\lambda + \mu)v = \lambda v + \mu v$

Remark 5. *The axioms for a ring and a vector space look similar because they share the same additive structure. The key difference is that a ring has an internal multiplication (elements of* R *multiply each other), whereas a vector space has an external scalar multiplication (elements of* V *are scaled by elements of a separate field* k*). A field is then just a ring where every nonzero element also has a multiplicative inverse, which is exactly what makes it a suitable set of scalars for a vector space.*

Remark 6. *We often say* abstract *vector space to emphasize that elements of* V *can be any objects (functions, tensors, matrices, etc.) satisfying these axioms — not just arrows or tuples in* $\mathbb{R}^n$.

Example (Non-standard vector space)**.** *Let $V = \{x \in \mathbb{R} : x > 0\}$ be the set of positive reals, with operations:*

$$u \oplus v := uv \quad (\text{“addition” is multiplication}), \qquad \lambda \odot u := u^\lambda \quad (\text{“scalar mult” is exponentiation}).$$

This looks nothing like a normal vector space, so we need to identify what the zero vector and additive inverses are before checking the axioms.
Zero vector. *We need $u \oplus 0_V = u$, i.e., $u \cdot 0_V = u$, so $0_V = 1$.*
Additive inverse. *We need $u \oplus (-u) = 0_V = 1$, i.e., $u \cdot (-u) = 1$, so $(-u) = 1/u$.*
Key axioms. *The non-obvious ones to check are:*

- *Distributivity over scalars: $(\lambda+\mu) \odot u = u^{\lambda+\mu} = u^\lambda \cdot u^\mu = (\lambda \odot u) \oplus (\mu \odot u)$.* ✓
- *Distributivity over vectors: $\lambda \odot (u \oplus v) = (uv)^\lambda = u^\lambda v^\lambda = (\lambda \odot u) \oplus (\lambda \odot v)$.* ✓
- *Scalar compatibility: $\lambda \odot (\mu \odot u) = \lambda \odot (u^\mu) = (u^\mu)^\lambda = u^{\lambda\mu} = (\lambda\mu) \odot u$.* ✓

All 10 axioms hold, so V is a vector space. The reason this works is that V is really just $\mathbb{R}$ in disguise: the map $\varphi : \mathbb{R} \to V$, $x \mapsto e^x$ converts ordinary addition into multiplication and ordinary scalar multiplication into exponentiation.

3.1 Vectors and Linear Combinations

Vector operations in $\mathbb{R}^n$ are defined component-wise:

- Addition: $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}$.
- Scalar multiplication: $\lambda \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \lambda x_1 \\ \vdots \\ \lambda x_n \end{pmatrix}$, where $\lambda \in \mathbb{R}$.

Definition 20 (Linear combination)**.** *A linear combination of $v_1, \ldots, v_n$ is $v = \lambda_1 v_1 + \cdots + \lambda_n v_n$ for scalars $\lambda_i \in k$.*

Example (Computing a linear combination)**.** *Let $v_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$.*

Express $v = \begin{pmatrix} 7 \\ 3 \end{pmatrix}$ as a linear combination of v_1 and v_2.
We need $\lambda_1, \lambda_2 \in \mathbb{R}$ such that $\lambda_1 v_1 + \lambda_2 v_2 = v$, i.e.,

$$\lambda_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 7 \\ 3 \end{pmatrix} \implies \begin{cases} \lambda_1 + 3\lambda_2 = 7 \\ 2\lambda_1 - \lambda_2 = 3. \end{cases}$$

From the second equation $\lambda_2 = 2\lambda_1 - 3$. Substituting: $\lambda_1 + 3(2\lambda_1 - 3) = 7 \implies 7\lambda_1 = 16 \implies \lambda_1 = \frac{16}{7}$, $\lambda_2 = \frac{11}{7}$. Hence

$$v = \tfrac{16}{7} v_1 + \tfrac{11}{7} v_2.$$

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3.2 Span and Standard Basis

3.2.1 Span

Definition 21 (Span)**.** *Let $A \subset \mathbb{R}^n$. The span of A is the set of all linear combinations of elements of A. If $A = \{v_1, \dots, v_k\}$, then*

$$\operatorname{span}(A) = \{\lambda_1 v_1 + \cdots + \lambda_k v_k : \lambda_i \in \mathbb{R}\}.$$

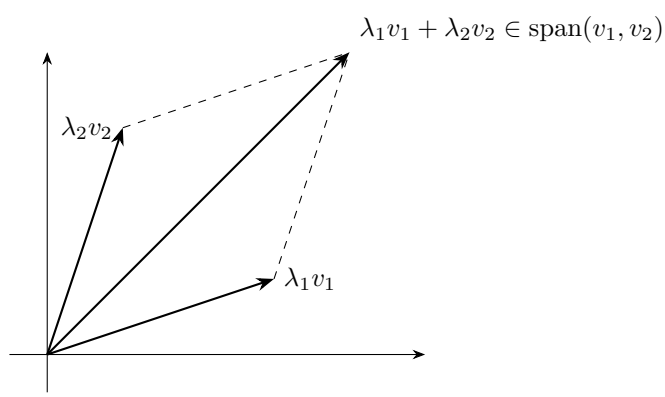
The span describes all points reachable by scaling and adding the given vectors: it can be a line, plane, or higher-dimensional subspace depending on the vectors.

Remark 7. *If $A = \{v\} \subset \mathbb{R}^2$ with $v \neq 0$, then $\operatorname{span}(v) = \{\lambda v : \lambda \in \mathbb{R}\}$ is the line through the origin in the direction of v.*

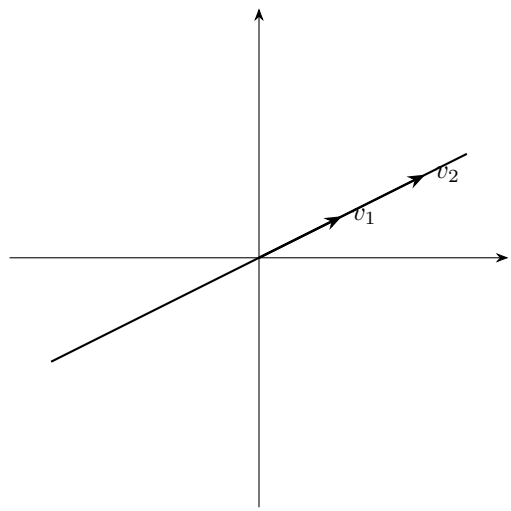
Example. $\operatorname{span}\begin{pmatrix}1\\1\end{pmatrix} = \left\{ t\begin{pmatrix}1\\1\end{pmatrix} \mid t \in \mathbb{R} \right\}$ *is a line in $\mathbb{R}^2$.*

Problem 5 (Span)**.** *Let $v_1 = \begin{pmatrix}3\\1\end{pmatrix}$, $v_2 = \begin{pmatrix}1\\3\end{pmatrix}$. Find $\operatorname{span}(v_1, v_2)$.*

For two linearly independent vectors $v_1, v_2 \in \mathbb{R}^2$, any vector in $\operatorname{span}(v_1, v_2)$ has the form $\lambda_1 v_1 + \lambda_2 v_2$:



If v_1 and v_2 are linearly dependent ($v_1 = \lambda v_2$), then $\operatorname{span}(v_1, v_2)$ is just a line:



Problem 6 (Span 2)**.** *Determine whether the first vector is in the span of the others. If so, write it as a linear combination.*

$$u = (2, 10, 7, 0), \quad u_1 = (3, 10, 7, 0), \quad u_2 = (1, 3, -2, 0), \quad u_3 = (2, 8, 1, 0); \quad u, u_i \in \mathbb{R}^4.$$

3.2.2 Span in $\mathbb{C}^n$

Definition 22 (Span over $\mathbb{C}$)**.**

$$\operatorname{span}_{\mathbb{C}}(v_1, \ldots, v_n) = \{\lambda_1 v_1 + \cdots + \lambda_n v_n : \lambda_i \in \mathbb{C}\}.$$

Example. *Let* $v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $v_2 = \begin{pmatrix} 0 \\ i \end{pmatrix} \in \mathbb{C}^2$. *Then* $\lambda_1 v_1 + \lambda_2 v_2 = \begin{pmatrix} \lambda_1 \\ i\lambda_2 \end{pmatrix}$. *Since* λ_1 *and* $i\lambda_2$ *can be any complex number,* $\operatorname{span}_{\mathbb{C}}(v_1, v_2) = \mathbb{C}^2$.

3.2.3 Standard Basis

Definition 23 (Standard basis of $\mathbb{R}^n$)**.** *The standard basis is* $\{e_1, \ldots, e_n\}$, *where* e_i *has a 1 in the* i*-th coordinate and 0 elsewhere:*

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \ldots, \quad e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

Equivalently, $e_1, \ldots, e_n$ *are the columns of* I_n.

Proposition 1. *Every* $x \in \mathbb{R}^n$ *can be written uniquely as* $x = x_1 e_1 + \cdots + x_n e_n$.[9]

[9] To verify the standard basis is indeed a basis, see Appendix (10.1).

The standard basis also clarifies the difference between real and complex vector spaces — the same vectors generate very different spans depending on the scalar field.

Example (Real vs complex span)**.** *Using e_1, e_2 in $\mathbb{C}^2$: over field $\mathbb{C}$, $\operatorname{span}_{\mathbb{C}}(e_1, e_2) = \mathbb{C}^2$ with $\dim_{\mathbb{C}} \mathbb{C}^2 = 2$. Over field $\mathbb{R}$, e_1, e_2 cannot produce vectors with imaginary parts. To span $\mathbb{C}^2$ over $\mathbb{R}$, we also need ie_1, ie_2, so $\operatorname{span}_{\mathbb{R}}(e_1, e_2, ie_1, ie_2)$, giving*

$$\dim_{\mathbb{R}} \mathbb{C}^2 = 4.$$

3.3 Examples of Vector Spaces

14 January, 2026

3.3.1 Coordinate Spaces

Definition 24 (Coordinate space)**.**

$$k^n = \left\{ \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \mid x_i \in k \right\}, \quad \dim k^n = n.$$

Standard basis: $\{e_1, \ldots, e_n\}$.

3.3.2 Polynomial Spaces

Definition 25 (Polynomial space)**.**

$$P_n(k) = \{a_n x^n + \cdots + a_1 x + a_0 \mid a_i \in k\}, \quad \dim P_n(k) = n + 1.$$

Standard basis: $\{1, x, x^2, \ldots, x^n\}$, with $P_n(k) \cong k^{n+1}$. The isomorphism is simply the map that reads off the coefficients:

$$a_0 + a_1 x + \cdots + a_n x^n \longleftrightarrow \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{pmatrix}.$$

A polynomial is completely determined by its $n + 1$ coefficients, so it carries exactly the same information as a vector in k^{n+1}.

Example. *$1 + x^2 \in P_2(\mathbb{R})$, which has standard basis $\{1, x, x^2\}$ and $\dim P_2(\mathbb{R}) = 3$.*
$1 + ix^3 \in P_3(\mathbb{C})$, which has standard basis $\{1, x, x^2, x^3\}$ and $\dim P_3(\mathbb{C}) = 4$.

Remark 8. *The space of all polynomials $P_\infty(k) = \{a_0 + a_1 x + \cdots : a_i \in k\}$ is infinite-dimensional, and the spaces form a nested sequence*

$$P_0(k) \subseteq P_1(k) \subseteq \cdots \subseteq P_\infty(k).$$

3.3.3 Matrix Spaces

Definition 26 (Matrix space)**.**

$$M_n(k) = \left\{ \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \mid a_{ij} \in k \right\}, \quad \dim M_n(k) = n^2.$$

Standard basis: $\{e_{ij} : 1 \leq i, j \leq n\}$, where e_{ij} has a 1 in the (i, j)-th entry and 0 elsewhere.

Example. *In $M_2(\mathbb{R})$: $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = 1e_{11} + 2e_{12} + 3e_{21} + 4e_{22}$.*

3.3.4 Function Spaces

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Definition 27 (Function space)**.** *Let D be a set and k a field. Define*

$$F(D, k) = \{f : D \to k\},$$

with pointwise operations $(f + g)(x) = f(x) + g(x)$ and $(\lambda f)(x) = \lambda f(x)$. This is a vector space over k.

Finite case. If $D = \{1, \ldots, n\}$, define the Kronecker delta functions

$$\delta_i(j) = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Then $\{\delta_1, \ldots, \delta_n\}$ is a basis of $F(D, k)$ and every f can be written uniquely as $f = \sum_{i=1}^{n} f(i)\, \delta_i$. Hence $\dim F(D, k) = n$.

Infinite case. If D is infinite, $F(D, k)$ is infinite-dimensional: there is no finite spanning set. The natural candidate for a basis, $\{\delta_x : x \in D\}$, is linearly independent but fails to span because linear combinations must be finite — a general $f : D \to k$ can be nonzero at infinitely many points and cannot be expressed as a finite sum of δ_x's.

Example. *Let $D = \{1, 2, 3\}$, $k = \mathbb{R}$, and f with $f(1) = 2$, $f(2) = -1$, $f(3) = 3$. The standard basis is $\{\delta_1, \delta_2, \delta_3\}$ and $f = 2\delta_1 - \delta_2 + 3\delta_3$.*

3.4 Subspaces

January 19, 2026

3.4.1 Definition and the Subspace Criterion

Definition 28 (Subspace)**.** *A subspace of V is a subset $U \subseteq V$ that is itself a vector space over the same field k.*

Proposition 2 (Subspace criterion)**.** *$U \subseteq V$ is a subspace if and only if:*

1. $0 \in U$,
2. $u, v \in U \Rightarrow u + v \in U$,
3. $u \in U$, $\lambda \in k \Rightarrow \lambda u \in U$.

Remark 9 (Proving a vector space in practice)**.** *To prove that a set U is a vector space over k, you do* ***not*** *always need to verify all 10 axioms. If U is a subset of an already-known vector space V over the same field k, then 7 of the 10 axioms are* inherited automatically *from V. In that case, it suffices to verify only the 3 subspace criteria. The full 10-axiom verification is only necessary when constructing a completely new vector space with no ambient space to inherit from.*

Example (Proving a subspace with the 3-criteria)**.** *Let $V = \mathbb{R}^3$ and $U = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$. Prove U is a subspace.*
Since $U \subset \mathbb{R}^3$, we only check the 3 criteria.

1. ***Zero****: $0 + 0 + 0 = 0$, so $(0, 0, 0) \in U$.* ✓
2. ***Addition****: Let $u = (x_1, y_1, z_1)$, $v = (x_2, y_2, z_2) \in U$. Then*
$$(x_1 + x_2) + (y_1 + y_2) + (z_1 + z_2) = (x_1 + y_1 + z_1) + (x_2 + y_2 + z_2) = 0.$$
so $u + v \in U$. ✓
3. ***Scalar mult****: $(\lambda x) + (\lambda y) + (\lambda z) = \lambda(x + y + z) = 0$, so $\lambda u \in U$.* ✓

Hence U is a subspace of $\mathbb{R}^3$. Note that U is a plane through the origin.

Problem 7 (Subspace criterion)**.** *Let $A \in M_n(k)$. Prove that $U = \{x \in k^n : Ax = \vec{0}\}$ is a subspace (the null space or kernel of A).*

Problem 8 (Subspace criterion 2)**.** *Let $A \in M_n(k)$. Show that $U = \{Ax \mid x \in k^n\}$ is a subspace of k^n (the image of A).*

Problem 9 (Subspace criterion 3)**.** *Let $V = k$, viewed as a vector space over k. Show that the only subspaces are $\{0\}$ and V itself.*

Problem 10 (Subspace criterion 4)**.** *Let $v_1, \ldots, v_n \in V$. Show that* $\operatorname{span}(v_1, \ldots, v_n)$ *is a subspace of V.*

3.4.2 Membership in a Span

January 21, 2026

Problem 11 (Span membership)**.** *Let* $v = \begin{pmatrix} 3 \\ 5 \\ -5 \end{pmatrix}$, $v_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$, $v_2 = \begin{pmatrix} 2 \\ 5 \\ 4 \end{pmatrix}$, $v_3 = \begin{pmatrix} 1 \\ 3 \\ 6 \end{pmatrix}$. *Decide whether $v \in \operatorname{span}(v_1, v_2, v_3)$.*

Problem 12 (Span membership 2)**.** *Let $f = 3x^2 + 5x - 5$, $f_1 = x^2 + 2x + 1$, $f_2 = 2x^2 + 5x + 4$, $f_3 = x^2 + 3x + 6$. Decide whether $f \in \operatorname{span}(f_1, f_2, f_3)$.*

3.4.3 Operations on Subspaces

Let W be a vector space and $U, V \subseteq W$ subspaces.

Definition 29 (Intersection)**.**

$$U \cap V = \{w \in W \mid w \in U \text{ and } w \in V\}.$$

The intersection of subspaces is a subspace; the smallest possible is $\{0\}$*.*

Definition 30 (Union)**.**

$$U \cup V = \{w \in W \mid w \in U \text{ or } w \in V\}.$$

The union of subspaces is not *necessarily a subspace.*

Proposition 3. *$U \cup V$ is a subspace of W if and only if $U \subseteq V$ or $V \subseteq U$.*

Proof. If $U \subseteq V$, then $U \cup V = V$, a subspace. Conversely, if neither contains the other, pick $u \in U \setminus V$ and $v \in V \setminus U$. If $U \cup V$ were a subspace, $u+v \in U \cup V$. If $u + v \in U$, then $v = (u + v) - u \in U$, a contradiction. If $u + v \in V$, then $u = (u + v) - v \in V$, also a contradiction. □

Definition 31 (Sum of subspaces)**.**

$$U + V = \{u + v \mid u \in U,\, v \in V\}.$$

Proposition 4. *If U and V are subspaces of W, then $U + V$ is a subspace of W.*

Proof. Apply the subspace criterion: $0 = 0 + 0 \in U + V$; if $w_1 = u_1 + v_1$ and $w_2 = u_2 + v_2$ then $w_1 + w_2 = (u_1 + u_2) + (v_1 + v_2) \in U + V$; for $\lambda \in K$, $\lambda w = \lambda u + \lambda v \in U + V$. □

Proposition 5. *$U + V = \operatorname{span}(U \cup V)$, i.e., $U + V$ is the smallest subspace containing both U and V.*

Proof. ($\subseteq$): Any $u + v \in U + V$ with $u, v \in U \cup V$ lies in $\operatorname{span}(U \cup V)$.
($\supseteq$): Any $w \in \operatorname{span}(U \cup V)$ can be grouped as a sum of a U-part and a V-part, so $w \in U + V$. □

3.4.4 Direct Sums

January 23, 2026

Definition 32 (Direct sum)**.** *We say U and V are in direct sum if $U \cap V = \{0\}$, and write $U \oplus V$ for their sum. We write*

$$W = U \oplus V \iff \begin{cases} W = U + V, \\ U \cap V = \{0\}. \end{cases}$$

Proof. $W = U \oplus V$ iff every $w \in W$ decomposes *uniquely* as $w = u + v$. Uniqueness follows since $u_1 - u_2 = v_2 - v_1 \in U \cap V = \{0\}$ forces $u_1 = u_2$. Conversely, if decompositions are unique and $w \in U \cap V$, then $0 = w + (-w) = 0 + 0$ forces $w = 0$. □

Analogy with sets.

Sets	Vector spaces
$A \cap B$	$U \cap V$
$A \cup B$	$U + V$
$A \sqcup B,\ A \cap B = \emptyset$	$U \oplus V,\ U \cap V = \{0\}$

Direct sums in function spaces.

Definition 33 (Support on a subset)**.** *Let $A \subseteq D$. Define $U = \{f \in F(D, \mathbb{R}) : f(x) = 0\ \forall x \notin A\}$. Then $U \cong F(A, \mathbb{R})$ is a subspace of $F(D, \mathbb{R})$.*

Example. *Let $D = \{1, 2, 3, 4, 5\}$, $A = \{2, 4\}$. Then $U = \{f : f(1) = f(3) = f(5) = 0\}$ is a 2-dimensional subspace, with each $f \in U$ determined by $(f(2), f(4))$.*

4 Basis and Dimension

January 26, 2026

4.1 Finite Dimensional Spaces

Let V be a vector space over a field k. Either $V = \{0\}$, or there exists $v_1 \neq 0$ in V. In the latter case, either $V = \operatorname{span}(v_1)$, or there exists $v_2 \notin \operatorname{span}(v_1)$, and so on. The maximum number of linearly independent vectors is the dimension of V.

Definition 34 (Finite dimensional vector space)**.** *V is finite-dimensional if $\exists$ a finite set $v_1, \ldots, v_n \in V$ with $V = \operatorname{span}(v_1, \ldots, v_n)$. Its dimension is*

$$\dim_k V = \min\{n : V = \operatorname{span}_k(v_1, \ldots, v_n)\}.$$

Example ($\mathbb{C}$ as a real vector space)**.** $\dim_{\mathbb{R}} \mathbb{C} = 2$ *(basis $\{1, i\}$). For $M_n(\mathbb{C})$: $\dim_{\mathbb{C}} M_n(\mathbb{C}) = n^2$ and $\dim_{\mathbb{R}} M_n(\mathbb{C}) = 2n^2$.*

Example (P_∞ is infinite-dimensional)**.** *Since $x^{n+1} \notin P_n = \operatorname{span}\{1, x, \ldots, x^n\}$ for all n, the space $P_\infty(\mathbb{R})$ requires arbitrarily many basis vectors.*

Problem 13 (Infinite dimension of P_∞)**.** *Prove that P_∞ over any field k has infinite dimension.*

4.2 Linear Independence

Definition 35 (Linear independence)**.** *Vectors $v_1, \ldots, v_n$ are linearly independent (LI) if*

$$\lambda_1 v_1 + \cdots + \lambda_n v_n = 0 \implies \lambda_1 = \cdots = \lambda_n = 0.$$

Definition 36 (Linear dependence)**.** *Linearly dependent (LD) is the negation of LI: there exist scalars, not all zero, with $\lambda_1 v_1 + \cdots + \lambda_n v_n = 0$.*

Example. $e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ *are LI:* $xe_1 + ye_2 = \begin{pmatrix} x \\ y \end{pmatrix} = 0 \implies x = y = 0$.

Proposition 6. *$v_1, \dots, v_n$ are LD if and only if one vector lies in the span of the others.*

Proof. ($\Rightarrow$) If $\lambda_1 v_1 + \cdots + \lambda_n v_n = 0$ with some $\lambda_k \neq 0$, then $v_k = \frac{-1}{\lambda_k} \sum_{i \neq k} \lambda_i v_i$. ($\Leftarrow$) If $v_k = \sum_{i \neq k} \lambda_i v_i$, rearranging gives a nontrivial relation. □

Remark 10. *If any vector $v_i = 0$, the family is LD.*

Example. *For $v_1 = \begin{pmatrix} a_1 \\ 0 \\ 0 \end{pmatrix}$, $v_2 = \begin{pmatrix} b_1 \\ b_2 \\ 0 \end{pmatrix}$, $v_3 = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$, if $a_1, b_2, c_3 \neq 0$ then v_1, v_2, v_3 are LI (solved by back-substitution). If $c_3 = 0$, all vectors lie in a 2D subspace and are LD.*

January 28, 2026

Problem 14 (Linear independence)**.** *Let $v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$, $v_3 = \begin{pmatrix} 4 \\ 9 \\ 5 \end{pmatrix}$.*

Are these LI?

4.3 Basis

4.3.1 Definition and the Main Basis Theorem

Definition 37 (Basis)**.** *$(v_1, \dots, v_n)$ is a basis of V if it spans V and is linearly independent.*

Remark 11. *If V has a basis of n vectors, then $\dim V = n$.*

January 30, 2026

Example (Standard basis of $\mathbb{R}^3$)**.** *$\{e_1, e_2, e_3\}$ is a basis of $\mathbb{R}^3$: it spans (any $v = xe_1 + ye_2 + ze_3$) and is LI (setting to zero gives $x = y = z = 0$).*

Lemma 2. *For $(v_1, \dots, v_n) \subset \mathbb{R}^k$: if $n < k$ the vectors cannot span $\mathbb{R}^k$; if $n > k$ they cannot be LI.*

Lemma 3. *If $u_1, \dots, u_n$ span V and $v_1, \dots, v_m$ are LI in V, then $n \geq m$.*

Theorem 4 (Main Basis Theorem)**.** *$(v_1, \dots, v_n)$ in V are equivalent (these four properties are all secretly the same thing):*

1. ***Basis**: they span V and are LI.*
2. ***Minimal span**: they span V and no $m < n$ vectors span V.*
3. ***Maximal LI**: they are LI and no $m > n$ vectors are LI.*
4. ***Unique coordinates**: every $v \in V$ writes uniquely as $v = \sum_{i=1}^n \lambda_i v_i$.*

Corollary 4.1 (Coordinate vectors)**.** *For basis $(v_1, \dots, v_n)$ of V, every $v = x_1 v_1 + \cdots + x_n v_n$ uniquely, and $[v]_e = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ is the coordinate vector of v.*

Corollary 4.2. *Any two bases of V have the same number of vectors. Hence $\dim V$ is well-defined.*

Proof. If $(e_1, \ldots, e_n)$ and $(f_1, \ldots, f_m)$ are both bases, the two-way spanning/independence argument gives $n \geq m$ and $m \geq n$. $\square$

February 2, 2026

4.3.2 Finding a Basis

There are several methods depending on how the space is described.

Method 1: Extracting from a spanning set.

Use this when you are given explicit vectors in k^m and want to find which ones form a basis.

1. Form a matrix A whose *columns* are the vectors.
2. Row reduce A to echelon form.
3. The original vectors corresponding to pivot columns form a basis.

Problem 15 (Finding a Basis)**.** *Let V be the subspace of $\mathbb{R}^4$ spanned by $v_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 1 \end{pmatrix}$, $v_2 = \begin{pmatrix} 2 \\ 1 \\ 3 \\ 2 \end{pmatrix}$, $v_3 = \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}$, $v_4 = \begin{pmatrix} 2 \\ 8 \\ 2 \\ 10 \end{pmatrix}$. Find a subset forming a basis of V.*

Method 2: Testing linear independence.

Use this when the vectors live in an abstract space (matrices, polynomials) that you can flatten into coordinates.

1. Map each vector to its coordinate vector in k^n using a known isomorphism.
2. Set up $a_1 v_1 + \cdots + a_n v_n = 0$ and solve.
3. Drop any vector whose coefficient is forced to be free (it is a linear combination of the others).

Problem 16 (Finding a Basis 2)**.** *Find a basis of the span of $M_1 = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$, $M_2 = \begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix}$, $M_3 = \begin{pmatrix} 4 & 5 \\ 5 & 7 \end{pmatrix}$ in $M_2(\mathbb{R})$.*

Method 3: Constructing from the general form.

Use this when the subspace is defined by constraints (e.g. symmetry, zero entries). Write out the most general element satisfying the constraints and identify the free parameters — each gives one basis vector.

Problem 17 (Finding a Basis 3)**.** *(1) Show $A \in M_2(\mathbb{C})$ is Hermitian iff $A = \begin{pmatrix} x & z \\ \bar{z} & y \end{pmatrix}$ for $x, y \in \mathbb{R}$, $z \in \mathbb{C}$.*
(2) Find a basis of the set U of all Hermitian matrices in $M_2(\mathbb{C})$, viewed as a real vector space.

Method 4: Bases of function spaces.

Use this when the space is $F(X)$ for a finite set X. The dimension equals the number of free values, and you verify a candidate basis by checking LI (evaluate at each point) and count.

Problem 18 (Finding a Basis 4)**.** *Let $X = \{-2, -1, 0, 1, 2\}$, $U = \{f \in F(X) : f(-x) = f(x)\ \forall x\}$. With $g_1(x) = 1$, $g_2(x) = \frac{1}{4}x^2(1 - x^2)$, $g_3(x) = \frac{1}{3}x^2(4 - x^2)$, prove $S = (g_1, g_2, g_3)$ is a basis of U.*

4.3.3 Proving a Set Forms a Basis

Method 1 (Dimension test)**.** *$(v_1, \ldots, v_n)$ is a basis of V iff $\dim(\operatorname{span}(v_1, \ldots, v_n)) = n$.*

Proposition 7 (Basis criterion)**.** *If $\dim V = n$, then for $v_1, \ldots, v_n \in V$, the following are equivalent: (1) basis, (2) spans V, (3) LI.*

Proof. A spanning set of $n = \dim V$ vectors is minimal, hence LI. An independent set of n vectors is maximal, hence spans V. □

Principle 1 (Transfer/Exchange principle)**.** *If $u_1, \ldots, u_n$ span V and $v_1, \ldots, v_n$ are LI, then both lists are bases of V.*

Example. *In P_n: $(1, x, \ldots, x^n)$ spans and $((x-a)^0, \ldots, (x-a)^n)$ is LI, so both are bases of P_n.*

February 11, 2026

4.3.4 Basis Extension and Extraction

Theorem 5 (Basis extension)**.** *Let V be finite-dimensional and $v_1, \ldots, v_m$ LI in V. Then they can be extended to a basis of V.*

Proof. Set $U = \operatorname{span}(v_1, \ldots, v_m)$. If $U = V$, done. Otherwise pick $v_{m+1} \in V \backslash U$; this gives a larger LI set. Repeat until the span equals V, which happens after at most $\dim V - m$ steps. □

Theorem 6 (Basis extraction)**.** *If $v_1, \ldots, v_n$ span V, one can extract a basis from this list.*

Problem 19 (Basis Extension)**.** *Let V be a finite-dimensional vector space and $L \subseteq V$ linearly independent. Show there is a basis $B \supseteq L$ of V.*

February 4–6, 2026

4.4 Coordinates and Change of Basis

4.4.1 Coordinate Vectors

If $(e_1, \dots, e_n)$ is a basis of V, every $v \in V$ writes uniquely as $v = \lambda_1 e_1 + \cdots + \lambda_n e_n$, and

$$[v]_e = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix}.$$

Example (Finding a coordinate vector)**.** *Find the coordinate vector of* $v = \begin{pmatrix} 5 \\ 3 \\ 4 \end{pmatrix}$ *in the basis* $f = (f_1, f_2, f_3)$ *where* $f_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$, $f_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, $f_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$.
We need x, y, z *such that* $xf_1 + yf_2 + zf_3 = v$. *Writing this as a system:*

$$\begin{pmatrix} 1 & 1 & 0 \\ -1 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 5 \\ 3 \\ 4 \end{pmatrix}.$$

Row reduce the augmented matrix:

$$\begin{pmatrix} 1 & 1 & 0 & 5 \\ -1 & 1 & 1 & 3 \\ 0 & 0 & 1 & 4 \end{pmatrix} \xrightarrow{R_2+R_1} \begin{pmatrix} 1 & 1 & 0 & 5 \\ 0 & 2 & 1 & 8 \\ 0 & 0 & 1 & 4 \end{pmatrix} \xrightarrow{R_2-R_3} \begin{pmatrix} 1 & 1 & 0 & 5 \\ 0 & 2 & 0 & 4 \\ 0 & 0 & 1 & 4 \end{pmatrix}.$$

Back-substitution: $z = 4$, $y = 2$, $x = 3$. *Therefore* $[v]_f = \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix}$.

Problem 20 (Coordinate vectors)**.** *Let* $X = \{-2, -1, 0, 1, 2\}$, $U = \{f \in F(X) : f(-x) = f(x)\}$, *with basis* $S = (g_1, g_2, g_3)$ *as above. Find* $[g]_S$ *for* g *with* $g(\pm 2) = 3$, $g(\pm 1) = g(0) = 2$.

Problem 21 (Coordinate vectors 2)**.** *Let* $D = \{1, 2, 3\}$. *Find the coordinates of* $f(x) = 2x^2 - 3$ *in the standard basis of* $F(D)$.

4.4.2 Direct Sum Decompositions and Dimension Formulas

Theorem 7 (Basis concatenation)**.** *If* $V = U \oplus U'$ *with bases* $(e_1, \dots, e_n)$ *and* $(f_1, \dots, f_m)$ *respectively, then* $(e_1, \dots, e_n, f_1, \dots, f_m)$ *is a basis of* V *and* $\dim(U \oplus U') = \dim U + \dim U'$.

Proof. LI: $\sum a_i e_i + \sum b_j f_j = 0$ gives $w = \sum a_i e_i = -\sum b_j f_j \in U \cap U' = \{0\}$, forcing all $a_i = b_j = 0$. Span: every $v \in U + U'$ writes $v = u + u'$. □

Theorem 8 (Dimension of sum)**.**

$$\dim(U + U') = \dim U + \dim U' - \dim(U \cap U').$$

Corollary 8.1. *If $U \subseteq V$, then $\dim U \leq \dim V$.*

Corollary 8.2. *If $U \subseteq V$, then U admits a complement: $\exists U' \subseteq V$ with $V = U \oplus U'$ and $\dim V = \dim U + \dim U'$.*

5 Linear Transformations

5.1 Functions: Basic Definitions

Let U and V be sets. A transformation $T : U \to V$ assigns to each $u \in U$ a unique element $T(u) \in V$. The set U is the domain, V the codomain, and $\mathrm{Im}(T) = \{T(u) : u \in U\} \subseteq V$ is the image.

Definition 38 (Injective, surjective, bijective)**.**

- *T is **surjective** if $\mathrm{Im}(T) = V$.*
- *T is **injective** if $T(u) = T(u') \implies u = u'$.*
- *T is **bijective** if it is both injective and surjective.*

Example. *$f : \mathbb{R} \to \mathbb{R}$, $x \mapsto x^3$ is bijective. $f : \mathbb{R} \to \mathbb{R}$, $x \mapsto x^2$ is neither injective nor surjective.*

Example. *$z \mapsto z^2$ on $\mathbb{C}$: not injective since $(-1)^2 = 1^2$, but surjective since every $w \in \mathbb{C}$ has a square root.*

5.1.1 Composition and Inverses

Given $T : U \to V$ and $S : V \to W$, the composition $R = S \circ T : U \to W$ is defined by $R(u) = S(T(u))$.

Example. *$g = \sin \circ (x^2) : \mathbb{R} \xrightarrow{x^2} \mathbb{R} \xrightarrow{\sin} \mathbb{R}$, so $g(x) = \sin(x^2)$.*

Proposition 8. *$T : U \to V$ is bijective $\iff$ T has an inverse $S : V \to U$ with $ST = \mathrm{id}_U$ and $TS = \mathrm{id}_V$.*

5.2 Linear Maps

Definition 39 (Linear transformation)**.** *Let U, V be vector spaces over k. A map $T : U \to V$ is **linear** if:*

1. *Additive: $T(u + u') = T(u) + T(u')$,*
2. *Scalar: $T(\lambda u) = \lambda T(u)$ for all $\lambda \in k$.*

5.2.1 Main Examples of Linear Maps

Matrix transformations. The primary example is $T_A : \mathbb{R}^n \to \mathbb{R}^n$, $T_A(x) = Ax$ for a fixed $A \in M_n(\mathbb{R})$:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \implies T_A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + 2y \\ 3x + 4y \end{pmatrix}.$$

Scaling.

Definition 40 (Scaling transformation)**.** *$T : U \to U$ is called scaling if there exists a basis $(u_1, \dots, u_n)$ and scalars $\lambda_1, \dots, \lambda_n \in k$ such that $T(u_i) = \lambda_i u_i$ for all i.*

The derivative.

$$D : P_n \to P_n, \quad f \mapsto f', \quad D(x^k) = kx^{k-1}.$$

D is linear: $(f+g)' = f' + g'$ and $(\lambda f)' = \lambda f'$. D is not injective ($D(1) = 0$) and not surjective ($x^n \notin \mathrm{Im}(D)$ since every polynomial in $\mathrm{Im}(D)$ has degree $\leq n-1$).

The indefinite integral.

$$\int : P_\infty \to P_\infty \quad \text{[linear, injective, not surjective]}.$$

Injective: if $\int f = 0$ then $f = 0$. Not surjective: no $f \in P_\infty$ satisfies $\int f = 1$.

Projection. Suppose $V = U \oplus U'$. Every $v \in V$ writes uniquely as $v = u + u'$. Define $P(v) := u$.

Proposition 9. *The projection $P : V \to U$ along U' is linear.*

Proof. Write $v_1 = u_1 + u_1'$ and $v_2 = u_2 + u_2'$. Then $v_1 + v_2 = (u_1 + u_2) + (u_1' + u_2')$, so $P(v_1 + v_2) = u_1 + u_2 = P(v_1) + P(v_2)$. Scalar case is analogous. □

5.2.2 Defining a Linear Map by its Action on a Basis

Proposition 10. *Let $(u_1, \dots, u_n)$ be a basis of U and $v_1, \dots, v_n$ arbitrary vectors in V. There exists a unique linear $T : U \to V$ with $T(u_i) = v_i$.*

Proof. Existence: for any $u = \lambda_1 u_1 + \cdots + \lambda_n u_n$, define $T(u) = \lambda_1 v_1 + \cdots + \lambda_n v_n$. Linearity follows from uniqueness of coordinates.
Uniqueness: any linear T with $T(u_i) = v_i$ is forced on all u by linearity. □

Proposition 11. *If S, T are linear, then $S \circ T$ is linear.*

5.3 Kernel and Image

February 18, 2026

Definition 41 (Kernel and image)**.**

$$\ker(T) = \{u \in U : T(u) = 0\}, \qquad \mathrm{Im}(T) = \{T(u) : u \in U\} \subseteq V.$$

Proposition 12. $\ker(T) \subseteq U$ *and* $\mathrm{Im}(T) \subseteq V$ *are subspaces.*

Proposition 13. *T is injective* $\iff$ $\ker(T) = \{0\}$.

Proof. ($\Leftarrow$) If $\ker(T) = \{0\}$ and $T(u) = T(u')$, then $T(u-u') = 0$, so $u-u' = 0$. ($\Rightarrow$) Contrapositive: if $\exists u \neq 0$ with $T(u) = 0$, then $T(u) = T(0)$ with $u \neq 0$. $\square$

Remark 12. *The kernel is only useful for linear T. For example,* $\ker(\sin) = \{k\pi : k \in \mathbb{Z}\}$ *is not a subspace.*

Proposition 14. *If $(e_1, \dots, e_n)$ is a basis of U, then* $\operatorname{Im}(T) = \operatorname{span}(T(e_1), \dots, T(e_n))$.

Proof. Any $v = T(u) = T(\lambda_1 e_1 + \cdots + \lambda_n e_n) = \lambda_1 T(e_1) + \cdots + \lambda_n T(e_n) \in \operatorname{span}(T(e_i))$. $\square$

5.3.1 Rank-Nullity Theorem

Theorem 9 (Rank-Nullity Theorem)**.** *Let $T : U \to V$ be a linear map with U finite-dimensional. Then*

$$\dim \ker(T) + \dim \operatorname{Im}(T) = \dim U.$$

Proof. Let $n = \dim U$, $k = \dim \ker(T)$, and let $(u_1, \dots, u_k)$ be a basis of $\ker(T)$. By basis extension, extend to a basis $(u_1, \dots, u_k, u_{k+1}, \dots, u_n)$ of U. We claim $(T(u_{k+1}), \dots, T(u_n))$ is a basis of $\operatorname{Im}(T)$.

Span: Any $T(u) = T(\sum \lambda_i u_i) = \sum_{i=k+1}^n \lambda_i T(u_i)$ since $T(u_i) = 0$ for $i \leq k$.

LI: Suppose $\sum_{i=k+1}^n \lambda_i T(u_i) = 0$. Then $T\big(\sum_{i=k+1}^n \lambda_i u_i\big) = 0$, so $\sum_{i=k+1}^n \lambda_i u_i \in \ker(T) = \operatorname{span}(u_1, \dots, u_k)$. Since $(u_1, \dots, u_n)$ is LI, all $\lambda_i = 0$.

Hence $\dim \operatorname{Im}(T) = n - k$, which gives the result. $\square$

Remark 13. $\dim \operatorname{Im}(T)$ *is called the* ***rank*** *of T and* $\dim \ker(T)$ *the* ***nullity****. For a matrix $A \in M_{m\times n}(k)$, rank-nullity says: rank(A) + nullity$(A) = n$ (number of columns).*

5.3.2 Examples of Kernel and Image

Example. *For the projection $P : V \to U$ along U':* $\operatorname{Im}(P) = U$ *and* $\ker(P) = U'$.

Example. $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $T_A : \mathbb{R}^2 \to \mathbb{R}^2$. *Solving* $\begin{cases} x + 2y = 0 \\ 3x + 4y = 0 \end{cases}$ *gives* $x = y = 0$, *so* $\ker(T_A) = \{0\}$ *and T_A is injective.*

Example. $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$. *Row reduction gives* $\ker(T_A) = \operatorname{span}\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$.

February 20, 2026

5.4 Matrix of a Linear Transformation

Definition 42 (Matrix of T in basis e)**.** *Let $e = (e_1, \dots, e_n)$ be a basis of U. The matrix of $T : U \to U$ in basis e is*

$$[T]_e = \big([T(e_1)]_e, \dots, [T(e_n)]_e\big).$$

The columns are the coordinate vectors of the images of the basis vectors.

Example. *For $T = T_A$ on $\mathbb{R}^2$ with the standard basis, $[T]_e = A$.*

Example (Derivative matrix for P_2)**.** *Working in P_2 with basis $e = (1, x, x^2)$, every polynomial $a + bx + cx^2$ has coordinate vector $(a, b, c)^T$. To build $[D]_e$, apply D to each basis vector and write the result's coordinates as a column:*

$$D(1) = 0 \to \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad D(x) = 1 \to \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad D(x^2) = 2x \to \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix},$$

so $[D]_e = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$. *To verify: $f = 3 + 5x - 2x^2$ has $f' = 5 - 4x$, and indeed*

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ 5 \\ -2 \end{pmatrix} = \begin{pmatrix} 5 \\ -4 \\ 0 \end{pmatrix} \longleftrightarrow 5 - 4x. \checkmark$$

Example (Matrix of T in a non-standard basis)**.** *Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3x + y \\ x + 3y \end{pmatrix}$, and let $f = (f_1, f_2)$ with $f_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $f_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. Find $[T]_f$.*

Column 1. $T(f_1) = \begin{pmatrix} 4 \\ 4 \end{pmatrix} = 4f_1 + 0f_2$, *so* $[T(f_1)]_f = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$.

Column 2. $T(f_2) = \begin{pmatrix} 2 \\ -2 \end{pmatrix} = 0f_1 + 2f_2$, *so* $[T(f_2)]_f = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$.

Therefore $[T]_f = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$, *which is diagonal. f_1 and f_2 are eigenvectors of T with eigenvalues 4 and 2.*

6 Isomorphisms and Coordinates

February 23, 2026

Definition 43 (Isomorphism)**.** *$T : U \to V$ is an isomorphism if it is linear and bijective.*

To check: verify T is linear, $\ker(T) = \{0\}$, and $\mathrm{Im}(T) = V$.

Theorem 10. *For $T : V \to V$ with matrix A in basis e, the following are equivalent:*

1. *T is an isomorphism.*
2. *A is invertible.*
3. $\det(A) \neq 0$.

Theorem 11. *$U \cong V \iff \dim U = \dim V$. In particular, every n-dimensional space is isomorphic to k^n.*

Example. *$P_8 \cong \mathbb{R}^9 \cong M_3(\mathbb{R})$ since all have dimension 9.*

6.1 The Canonical Isomorphism

Definition 44 (Coordinate map)**.** *Given vectors $v_1, \dots, v_n \in V$, define $\Phi : k^n \to V$ by*

$$\Phi \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} = \lambda_1 v_1 + \cdots + \lambda_n v_n.$$

Then $\operatorname{Im}(\Phi) = \operatorname{span}(v_1, \dots, v_n)$.

Theorem 12. *Φ is an isomorphism $\iff$ $(v_1, \dots, v_n)$ is a basis for V. In this case, $[v]_e = \Phi^{-1}(v)$.*

6.2 Change of Basis

February 27 – March 13, 2026

6.2.1 Change of Basis Matrix

Let $e = (e_1, \dots, e_n)$ and $f = (f_1, \dots, f_n)$ be two bases of V, and let M_e and M_f denote the matrices whose columns are the basis vectors of e and f written in standard coordinates. The change of basis matrices are:

- $Q = M_e^{-1} M_f$: converts f-coordinates to e-coordinates.
- $P = M_f^{-1} M_e = Q^{-1}$: converts e-coordinates to f-coordinates.

Remark 14. *When e is the standard basis, $M_e = I$, so the formulas simplify to $Q = M_f$ and $P = M_f^{-1}$.*

Example (Building Q when neither basis is standard)**.** *Let $e = \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right)$ and $f = \left(\begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right)$. Find Q.*

$$M_e = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad M_f = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad M_e^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}.$$

$$Q = M_e^{-1} M_f = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad P = Q^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}.$$

6.2.2 Change of Basis Formula

Theorem 13 (Change of basis)**.** *If $A = [T]_e$ and $B = [T]_f$, then*

$$B = PAQ, \quad P = Q^{-1}.$$

Remark 15 (The $Q^{-1}AQ$ shortcut)**.** *Since $P = Q^{-1}$, the formula is exactly $B = Q^{-1}AQ$. In practice: build Q (columns = new basis vectors, or $M_e^{-1} M_f$ in general), then compute $Q^{-1}AQ$.*

Example (Full change of basis computation)**.** *Let e be the standard basis of $\mathbb{R}^2$, $f_1 = \begin{pmatrix}1\\3\end{pmatrix}$, $f_2 = \begin{pmatrix}2\\5\end{pmatrix}$, and $T : \begin{pmatrix}x\\y\end{pmatrix} \mapsto \begin{pmatrix}2x-7y\\4x-3y\end{pmatrix}$. Find $[T]_f$.*

Step 1: $Q = \begin{pmatrix}1 & 2\\3 & 5\end{pmatrix}$.

Step 2: $P = Q^{-1} = \begin{pmatrix}-5 & 2\\3 & -1\end{pmatrix}$.

Step 3:

$$AQ = \begin{pmatrix}2 & -7\\4 & -3\end{pmatrix}\begin{pmatrix}1 & 2\\3 & 5\end{pmatrix} = \begin{pmatrix}-19 & -31\\-5 & -7\end{pmatrix}, \quad B = PAQ = \begin{pmatrix}85 & 141\\-52 & -86\end{pmatrix}.$$

7 Diagonalization

March 13–16, 2026

7.1 Eigenvalues and Eigenvectors

Definition 45 (Eigenvalue and eigenvector)**.** *$\lambda \in k$ is an eigenvalue of T if $\exists\, v \neq 0$ with $Tv = \lambda v$; v is an eigenvector. The eigenspace is*

$$E_\lambda = \{v \in V : Tv = \lambda v\}.$$

Proposition 15. *E_λ is a subspace of V.*

Proposition 16. *If $\lambda_1 \neq \lambda_2$, then $E_{\lambda_1} \cap E_{\lambda_2} = \{0\}$.*

Proof. If $v \in E_{\lambda_1} \cap E_{\lambda_2}$, then $\lambda_1 v = Tv = \lambda_2 v$, so $(\lambda_1 - \lambda_2)v = 0$. Since $\lambda_1 \neq \lambda_2$, we get $v = 0$. □

7.2 The Characteristic Polynomial

Definition 46 (Characteristic polynomial)**.**

$$\chi(\lambda) = \det(A - \lambda\,\mathrm{Id}).$$

Theorem 14. *λ is an eigenvalue of $A \iff \lambda$ is a root of χ.*

7.3 Algebraic and Geometric Multiplicity

March 18, 2026

Definition 47 (Algebraic and geometric multiplicity)**.** *For an eigenvalue λ of A:*

- *The **algebraic multiplicity** (AM) of λ is its exponent as a root of $\chi(\lambda)$.*
- *The **geometric multiplicity** (GM) of λ is $\dim E_\lambda = \dim\ker(A - \lambda\,\mathrm{Id})$.*

Theorem 15 (AM-GM inequality)**.** *For every eigenvalue λ:*

$$1 \leq \mathrm{GM}(\lambda) \leq \mathrm{AM}(\lambda).$$

A is diagonalizable $\iff \mathrm{GM}(\lambda) = \mathrm{AM}(\lambda)$ for every eigenvalue, i.e.,

$$V = E_{\lambda_1} \oplus E_{\lambda_2} \oplus \cdots \oplus E_{\lambda_k}.$$

Remark 16. *In particular, if A has n distinct eigenvalues (all $AM = 1$), then A is automatically diagonalizable.*

7.4 Diagonalization

Definition 48 (Diagonalizable)**.** *T (or its matrix A) is diagonalizable if $\exists$ a basis $e = (e_1, \ldots, e_n)$ and scalars $\lambda_1, \ldots, \lambda_n$ with $T(e_i) = \lambda_i e_i$.*

How to diagonalize A.

1. Compute $\chi(\lambda) = \det(A - \lambda \operatorname{Id})$ and find its roots (the eigenvalues).
2. For each eigenvalue λ_i, find $\ker(A - \lambda_i \operatorname{Id})$ to get eigenvectors.
3. If there are n linearly independent eigenvectors, write them as columns of Q; then $Q^{-1}AQ$ is diagonal.

Example (Diagonalization of a 4×4 matrix)**.** *Diagonalize* $A = \begin{pmatrix} 5 & -2 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & 4 \end{pmatrix}$.

Characteristic polynomial. *A is block diagonal:*

$$\chi(\lambda) = (5-\lambda)(3-\lambda) \cdot \big[(1-\lambda)(4-\lambda) - 4\big] = -\lambda(\lambda-3)(\lambda-5)^2.$$

Eigenvalues: $\lambda = 0$ (AM= 1), $\lambda = 3$ (AM= 1), $\lambda = 5$ (AM= 2).

Eigenvectors: *For $\lambda = 5$: $E_5 = \operatorname{span}\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 2 \end{pmatrix}$ (GM= 2=AM ✓). For $\lambda = 3$: $E_3 = \operatorname{span}\begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$. For $\lambda = 0$: $E_0 = \operatorname{span}\begin{pmatrix} 0 \\ 0 \\ -2 \\ 1 \end{pmatrix}$.*

$$P = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -2 \\ 0 & 2 & 0 & 1 \end{pmatrix}, \quad P^{-1}AP = \begin{pmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Example (3×3 diagonalization)**.** *Let* $A = \begin{pmatrix} 1 & 0 & 2 \\ 3 & 4 & 3 \\ 2 & 0 & 1 \end{pmatrix}$. *Expanding along the second column:*

$$\chi(\lambda) = (4-\lambda) \det \begin{pmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{pmatrix} = (\lambda-4)(\lambda-3)(\lambda+1).$$

Three distinct eigenvalues $\Rightarrow$ diagonalizable. Eigenvectors: $f_1 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, $f_2 = \begin{pmatrix} 1 \\ -6 \\ 1 \end{pmatrix}$, $f_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$.

$$Q = \begin{pmatrix} 0 & 1 & -1 \\ 1 & -6 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad A = QDQ^{-1}.$$

7.5 Similar Matrices

Definition 49 (Similar matrices)**.** *A and B are **similar** if there exists an invertible Q with $A = Q^{-1}BQ$.*

Problem 22 (Similar matrices have equal characteristic polynomials)**.** *Let A be similar to B, i.e., $A = Q^{-1}BQ$ for some invertible Q. Prove $\chi_A = \chi_B$.*

Problem 23 (Characteristic polynomial and invertibility)**.** *Let $A \in M_n(\mathbb{R})$ be non-invertible. Show $\chi(\lambda)$ has a root at $\lambda = 0$.*

8 Inner Product Spaces

March 18, 2026

An inner product space is a vector space V equipped with an inner product — a way to measure length, angle, and perpendicularity. The motivating example is $\mathbb{R}^n$ where $\|v\| = \sqrt{x_1^2 + \cdots + x_n^2}$.

Definition 50 (Inner product)**.** *Let V be a real vector space. An inner product on V is a map $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$ satisfying:*

1. ***Linearity on the left**: $\langle v_1 + v_2, w \rangle = \langle v_1, w \rangle + \langle v_2, w \rangle$ and $\langle \lambda v, w \rangle = \lambda \langle v, w \rangle$.*
2. ***Symmetry**: $\langle u, v \rangle = \langle v, u \rangle$.*
3. ***Positive definiteness**: $\langle v, v \rangle \geq 0$, and $\langle v, v \rangle = 0 \iff v = 0$.*

*A vector space with an inner product is called an **inner product space**.*

Remark 17. *By symmetry, linearity on the left implies linearity on the right: $\langle u, v_1 + v_2 \rangle = \langle u, v_1 \rangle + \langle u, v_2 \rangle$.*

Definition 51 (Norm)**.** *The norm induced by an inner product is $\|v\| = \sqrt{\langle v, v \rangle}$.*

Example (Standard inner product on $\mathbb{R}^n$)**.** *For $v = (x_1, \ldots, x_n)^T$ and $w = (y_1, \ldots, y_n)^T$, the dot product $\langle v, w \rangle = x_1 y_1 + \cdots + x_n y_n$ is an inner product.*

Example (Function space inner product)**.** *On $V = C^0([0,1])$, define $\langle f, g \rangle = \int_0^1 f(x)g(x)\,dx$. This satisfies all three axioms; positive definiteness uses the fact that if $f^2 \geq 0$ is continuous and $\int f^2 = 0$, then $f = 0$ everywhere.*

Example ($M_n(\mathbb{R})$)**.** *For $A = (a_{ij})$ and $B = (b_{ij})$, define $\langle A, B\rangle = \sum_{i,j} a_{ij}b_{ij}$ with $\|A\| = \sqrt{\sum_{i,j} a_{ij}^2}$.*

8.1 Norm and Pythagorean Theorem

Proposition 17 (Norm expansion)**.** *For any $u, v \in V$:*

$$\|u+v\|^2 = \|u\|^2 + 2\langle u, v\rangle + \|v\|^2.$$

Proof. $\langle u+v, u+v\rangle = \langle u, u\rangle + \langle u, v\rangle + \langle v, u\rangle + \langle v, v\rangle = \|u\|^2 + 2\langle u, v\rangle + \|v\|^2$. □

Theorem 16 (Pythagorean theorem)**.** $u \perp v \iff \langle u, v\rangle = 0 \iff \|u+v\|^2 = \|u\|^2 + \|v\|^2$.

8.2 Cauchy-Schwarz Inequality

Theorem 17 (Cauchy-Schwarz Inequality)**.** *For any u, v in an inner product space V:*

$$|\langle u, v\rangle| \leq \|u\|\,\|v\|,$$

with equality if and only if u and v are linearly dependent.

Proof. If $v = 0$ both sides are 0. Otherwise, let $w = u - \frac{\langle u,v\rangle}{\|v\|^2}v$ (the component of u orthogonal to v). Then $\langle w, v\rangle = 0$, and by positive definiteness:

$$0 \leq \|w\|^2 = \left\|u - \frac{\langle u, v\rangle}{\|v\|^2}v\right\|^2 = \|u\|^2 - \frac{|\langle u, v\rangle|^2}{\|v\|^2},$$

which rearranges to $|\langle u, v\rangle|^2 \leq \|u\|^2\|v\|^2$. □

Corollary 17.1 (Triangle inequality)**.** $\|u+v\| \leq \|u\| + \|v\|$.

Proof. $\|u+v\|^2 = \|u\|^2 + 2\langle u, v\rangle + \|v\|^2 \leq \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = (\|u\| + \|v\|)^2$. □

8.3 Complex Inner Products

Definition 52 (Inner product on $\mathbb{C}^n$)**.** *For $x, y \in \mathbb{C}^n$, define*

$$\langle x, y\rangle = \sum_{i=1}^{n} x_i\overline{y_i}.$$

Then $\|x\|^2 = \sum_{i=1}^{n} |x_i|^2 \geq 0$, with equality iff $x = 0$.

Remark 18. *For complex inner products the symmetry axiom becomes* ***conjugate symmetry****: $\langle u, v\rangle = \overline{\langle v, u\rangle}$. This means the inner product is only* conjugate-linear *on the right: $\langle u, \lambda v\rangle = \overline{\lambda}\langle u, v\rangle$.*

March 23, 2026

8.4 Orthonormal Bases

Definition 53 (Orthogonal and orthonormal)**.** *A set of vectors* $(e_1, \ldots, e_n)$ *is:*

- ***Orthogonal*** *if* $\langle e_i, e_j \rangle = 0$ *for all* $i \neq j$*.*
- ***Orthonormal*** *(ONB) if* $\langle e_i, e_j \rangle = \delta_{ij}$ *(orthogonal and each* $\|e_i\| = 1$*).*

Proposition 18. *Any orthogonal set of nonzero vectors is linearly independent.*

Proof. Suppose $\sum \lambda_i e_i = 0$. Taking the inner product with e_k:

$$0 = \left\langle e_k, \sum_i \lambda_i e_i \right\rangle = \lambda_k \|e_k\|^2,$$

so $\lambda_k = 0$ since $\|e_k\| \neq 0$. □

Proposition 19 (Coordinates in an ONB)**.** *If* $(e_1, \ldots, e_n)$ *is an ONB of* V*, then for any* $v \in V$*:*

$$v = \sum_{i=1}^{n} \langle v, e_i \rangle \, e_i.$$

In other words, the i*-th coordinate of* v *in the ONB is simply* $\langle v, e_i \rangle$*.*

Proof. Write $v = \sum \lambda_i e_i$. Taking $\langle \cdot, e_k \rangle$ of both sides: $\langle v, e_k \rangle = \lambda_k \|e_k\|^2 = \lambda_k$. □

March 25, 2026

8.5 Gram-Schmidt Orthonormalization

Given a linearly independent set, Gram-Schmidt produces an ONB with the same span by subtracting off the projection onto each previous vector.

Theorem 18 (Gram-Schmidt)**.** *Let* $(v_1, \ldots, v_n)$ *be linearly independent vectors in an inner product space* V*. Define:*

$$u_1 = \frac{v_1}{\|v_1\|},$$

and for $k = 2, \ldots, n$*:*

$$\tilde{u}_k = v_k - \sum_{j=1}^{k-1} \langle v_k, u_j \rangle \, u_j, \qquad u_k = \frac{\tilde{u}_k}{\|\tilde{u}_k\|}.$$

Then $(u_1, \ldots, u_n)$ *is an ONB of* $\operatorname{span}(v_1, \ldots, v_n)$*.*

Example (Gram-Schmidt in $\mathbb{R}^2$)**.** *Let* $v_1 = \begin{pmatrix} 5 \\ 0 \end{pmatrix}$ *and* $v_2 = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$*. Set* $u_1 = v_1/\|v_1\| = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$*.*

Subtract the projection of v_2 onto u_1:

$$\tilde{u}_2 = v_2 - \langle v_2, u_1 \rangle u_1 = \begin{pmatrix} 3 \\ 2 \end{pmatrix} - 3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}.$$

Then $u_2 = \tilde{u}_2 / \|\tilde{u}_2\| = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, and $\{u_1, u_2\}$ is the standard ONB.

Remark 19. *The key identity driving Gram-Schmidt is $k = \frac{\langle v_k, u_j \rangle}{\langle u_j, u_j \rangle} = \langle v_k, u_j \rangle$ (since $\|u_j\| = 1$). This is exactly the projection coefficient.*

8.6 Orthogonal Matrices

March 25, 2026

Definition 54 (Orthogonal matrix)**.** *A square matrix $A \in M_n(\mathbb{R})$ is orthogonal if its columns form an ONB of $\mathbb{R}^n$, equivalently if $A^T A = I$.*

Proposition 20. *The following are equivalent for a square matrix A:*

1. *A is orthogonal (columns form an ONB).*
2. *$A^T A = I$.*
3. *$AA^T = I$.*
4. *Rows of A form an ONB.*

In particular, an orthogonal matrix satisfies $A^{-1} = A^T$.

Proof. Write $A = (c_1, \ldots, c_n)$. Then $(A^T A)_{ij} = \langle c_i, c_j \rangle$, so $A^T A = I$ iff $\langle c_i, c_j \rangle = \delta_{ij}$, establishing (1) $\iff$ (2). Since $A^T A = I \implies \det(A) = \pm 1 \neq 0$, A is invertible, and $A^T = A^{-1}$ implies $AA^T = I$ as well, giving (3). The rows of A are the columns of A^T, and $AA^T = I$ says exactly that the rows are orthonormal, giving (4). □

Remark 20. *The change of basis formula simplifies nicely when Q is orthogonal: $B = Q^{-1}AQ = Q^T AQ$.*

8.7 Spectral Theorem for Symmetric Matrices

March 25, 2026

Proposition 21. *If A is symmetric (i.e., $A^T = A$) and $\lambda_1 \neq \lambda_2$ are distinct eigenvalues, then $E_{\lambda_1} \perp E_{\lambda_2}$.*

Proof. Let $v_1 \in E_{\lambda_1}$, $v_2 \in E_{\lambda_2}$. Then:

$$\lambda_1 \langle v_1, v_2 \rangle = \langle Av_1, v_2 \rangle = \langle v_1, A^T v_2 \rangle = \langle v_1, Av_2 \rangle = \lambda_2 \langle v_1, v_2 \rangle.$$

Since $\lambda_1 \neq \lambda_2$, we get $\langle v_1, v_2 \rangle = 0$. □

Theorem 19 (Spectral Theorem, Version 1)**.** *Every real symmetric matrix is diagonalizable.*

Theorem 20 (Spectral Theorem, Version 2)**.** *If A is symmetric, then there exists an ONB of $\mathbb{R}^n$ consisting of eigenvectors of A. Equivalently, there exists an orthogonal matrix Q and a diagonal matrix D such that*

$$A = QDQ^{-1} = QDQ^T.$$

Example. *Let $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$. The characteristic polynomial is $\chi(\lambda) = (\lambda - 3)(\lambda + 1)$, giving eigenvalues $\lambda_1 = 3$, $\lambda_2 = -1$.*

Eigenvectors: $v_3 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $v_{-1} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$. These are already orthogonal (as guaranteed by the spectral theorem), so after normalizing:

$$Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}, \quad A = QDQ^T.$$

9 Linear Transformations on Inner Product Spaces

Chapter 8

9.1 Linear Functionals

Definition 55 (Linear functional)**.** *A linear functional on V is a linear map $\varphi : V \to k$.*

Example. *On $\mathbb{R}^n$: fixing $w \in \mathbb{R}^n$, the map $\varphi_w(v) = \langle v, w \rangle$ is a linear functional. On P_n: evaluation $\varphi_a(f) = f(a)$ and integration $\varphi(f) = \int_0^1 f(x)\,dx$ are linear functionals.*

Theorem 21 (Riesz Representation Theorem)**.** *Let V be a finite-dimensional inner product space and $\varphi : V \to k$ a linear functional. Then there exists a unique $w \in V$ such that*

$$\varphi(v) = \langle v, w \rangle \quad \textit{for all } v \in V.$$

Proof. Let $(e_1, \ldots, e_n)$ be an ONB. Define $w = \sum_i \overline{\varphi(e_i)}\, e_i$. Then for any $v = \sum_j \langle v, e_j \rangle e_j$:

$$\langle v, w \rangle = \sum_j \langle v, e_j \rangle \varphi(e_j) = \varphi\left(\sum_j \langle v, e_j \rangle e_j \right) = \varphi(v).$$

Uniqueness: if $\langle v, w \rangle = \langle v, w' \rangle$ for all v, then $\langle v, w - w' \rangle = 0$ for all v, so taking $v = w - w'$ gives $\|w - w'\|^2 = 0$, hence $w = w'$. $\square$

9.2 Orthogonal Projections

Definition 56 (Orthogonal complement)**.** *For a subspace $U \subseteq V$, the orthogonal complement is*

$$U^\perp = \{v \in V : \langle v, u \rangle = 0 \ \forall u \in U\}.$$

Proposition 22. *$U^\perp$ is a subspace, and $V = U \oplus U^\perp$.*

Proof. Subspace: $\langle 0, u\rangle = 0$ for all u; sums and scalar multiples of vectors in $U^\perp$ are in $U^\perp$ by linearity. For the decomposition: given an ONB $(e_1, \ldots, e_k)$ of U, any $v \in V$ can be written as $v = u + u^\perp$ where

$$u = \sum_{i=1}^{k} \langle v, e_i\rangle e_i \in U, \qquad u^\perp = v - u.$$

One checks $u^\perp \in U^\perp$, and $U \cap U^\perp = \{0\}$ since $w \in U \cap U^\perp$ implies $\langle w, w\rangle = 0$. □

Definition 57 (Orthogonal projection)**.** *Given $V = U \oplus U^\perp$, the **orthogonal projection** onto U is the linear map $P_U : V \to U$ defined by $P_U(v) = u$ where $v = u + u^\perp$.*

If $(e_1, \ldots, e_k)$ is an ONB of U, then explicitly:

$$P_U(v) = \sum_{i=1}^{k} \langle v, e_i\rangle\, e_i.$$

Proposition 23. *The orthogonal projection P_U satisfies:*

1. *$P_U^2 = P_U$ (it is idempotent),*
2. *$\ker(P_U) = U^\perp$ and $\mathrm{Im}(P_U) = U$,*
3. *$P_U(v) = v$ for all $v \in U$.*

Example (Projection onto a line)**.** *Let $U = \mathrm{span}(u)$ in $\mathbb{R}^n$. Then $P_U(v) = \frac{\langle v,u\rangle}{\|u\|^2} u$. In terms of the unit vector $\hat{u} = u/\|u\|$: $P_U(v) = \langle v, \hat{u}\rangle \hat{u}$.*

Remark 21 (Best approximation)**.** *For a subspace $U \subseteq V$, the orthogonal projection $P_U(v)$ is the closest point in U to v: it minimizes $\|v - u\|$ over all $u \in U$. This is the geometric meaning of projection.*

9.3 The Adjoint of a Linear Operator

Definition 58 (Adjoint)**.** *Let $T : V \to V$ be a linear operator on a finite-dimensional inner product space. The **adjoint** of T is the unique linear operator $T^* : V \to V$ satisfying*

$$\langle T(u), v\rangle = \langle u, T^*(v)\rangle \quad \text{for all } u, v \in V.$$

Remark 22 (Existence and uniqueness)**.** *For each fixed v, the map $u \mapsto \langle T(u), v\rangle$ is a linear functional. By the Riesz representation theorem, there exists a unique $w \in V$ with $\langle T(u), v\rangle = \langle u, w\rangle$ for all u. Setting $T^*(v) = w$ defines T^*.*

Proposition 24 (Matrix of the adjoint)**.** *If $A = [T]_e$ in an ONB e, then $[T^*]_e = A^*$ (the conjugate transpose). For real matrices in an ONB, $[T^*]_e = A^T$.*

Proposition 25 (Properties of the adjoint)**.**

1. $(S+T)^* = S^* + T^*$
2. $(\lambda T)^* = \overline{\lambda} T^*$
3. $(ST)^* = T^* S^*$
4. $(T^*)^* = T$
5. *If T is invertible, $(T^{-1})^* = (T^*)^{-1}$*

9.4 Self-Adjoint, Unitary, and Normal Operators

Definition 59 (Self-adjoint, unitary, normal)**.** *Let $T : V \to V$ be a linear operator.*

- *T is **self-adjoint** (Hermitian) if $T^* = T$.*
- *T is **unitary** (orthogonal if real) if $T^*T = TT^* = \mathrm{Id}$, i.e., $T^* = T^{-1}$.*
- *T is **normal** if $T^*T = TT^*$.*

Remark 23. *Both self-adjoint and unitary operators are normal. In matrix terms over an ONB:*

- *Self-adjoint $\leftrightarrow A = A^*$ (symmetric if real).*
- *Unitary $\leftrightarrow A^*A = I$ (orthogonal if real).*
- *Normal $\leftrightarrow A^*A = AA^*$.*

Proposition 26 (Eigenvalues of self-adjoint operators are real)**.** *If T is self-adjoint and $Tv = \lambda v$ with $v \neq 0$, then $\lambda \in \mathbb{R}$.*

Proof. $\lambda \|v\|^2 = \langle Tv, v\rangle = \langle v, T^*v\rangle = \langle v, Tv\rangle = \overline{\lambda}\|v\|^2$. Since $\|v\|^2 > 0$, we get $\lambda = \overline{\lambda}$, so $\lambda \in \mathbb{R}$. □

Proposition 27 (Eigenvalues of unitary operators have modulus 1)**.** *If T is unitary and $Tv = \lambda v$ with $v \neq 0$, then $|\lambda| = 1$.*

Proof. $\|v\|^2 = \langle v, v\rangle = \langle T^*Tv, v\rangle = \langle Tv, Tv\rangle = |\lambda|^2\|v\|^2$. Hence $|\lambda|^2 = 1$. □

9.5 Spectral Theorem (General Version)

Theorem 22 (Spectral Theorem for Normal Operators over $\mathbb{C}$)**.** *Let V be a finite-dimensional complex inner product space and $T : V \to V$ normal. Then V has an ONB consisting of eigenvectors of T.*

Theorem 23 (Spectral Theorem for Self-Adjoint Operators over $\mathbb{R}$)**.** *Let V be a finite-dimensional real inner product space and $T : V \to V$ self-adjoint. Then V has an ONB consisting of eigenvectors of T.*

In matrix terms, the spectral theorem says: A is a real symmetric (or complex normal) matrix if and only if $A = QDQ^{-1}$ where Q is orthogonal (unitary) and D is real diagonal.

Algorithm for orthogonal diagonalization.

1. Find all eigenvalues (roots of χ).
2. For each eigenvalue, find a basis of the eigenspace.
3. If eigenspaces have dimension > 1 (repeated eigenvalue), apply Gram-Schmidt within that eigenspace to get an orthogonal basis.
4. Normalize all eigenvectors. Eigenvectors from different eigenspaces are automatically orthogonal.
5. Assemble the normalized eigenvectors as columns of Q.

Example (Orthogonal diagonalization)**.** *Let* $A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$. *Since A is symmetric, it is orthogonally diagonalizable.*

Characteristic polynomial:

$$\chi(\lambda) = -(\lambda - 1)^2(\lambda - 4).$$

Eigenvalues: $\lambda_1 = 1$ *(AM* $= 2$*) and* $\lambda_2 = 4$ *(AM* $= 1$*).*

For $\lambda = 4$***:*** $\ker(A - 4I)$ *gives* $v = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$. *Normalized:* $e_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

For $\lambda = 1$***:*** $\ker(A - I)$ *gives the 2D space* $\{x_1 + x_2 + x_3 = 0\}$*, with basis* $\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$. *Apply Gram-Schmidt to get an orthogonal basis, then normalize to get* e_1, e_2.

Result:

$$Q = (e_1 \mid e_2 \mid e_3), \quad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \quad A = QDQ^T.$$

10 Appendix

10.1 Proof: Standard Basis is a Basis

Claim 1. $e_1, \ldots, e_n$ span $\mathbb{R}^n$.

Proof. $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = x_1 e_1 + \cdots + x_n e_n.$ □

Claim 2. $e_1, \ldots, e_n$ are linearly independent.

Proof. $\lambda_1 e_1 + \cdots + \lambda_n e_n = 0 \implies \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} = 0 \implies \lambda_i = 0.$ □

11 Solutions

Solution 1 (Hermitian Matrix)**.** $(AB)^* = (\overline{AB})^T = (\overline{A}\,\overline{B})^T = \overline{B}^T \overline{A}^T = B^* A^*$.

Solution 2 (Hermiticity and Anti-Hermiticity)**.** *($\Rightarrow$): Suppose A is Hermitian, then $A = \overline{A}^t$, then $(iA)^* = \overline{iA^t} = -iA^* = -iA$. Therefore, $-(iA)^* = iA$.*
($\Leftarrow$): Suppose $iA = -(iA)^*$, then $iA = -(iA)^* = -(-iA^*) = iA^* = iA$. Therefore, $iA^* = iA \implies A = A^*$.

Solution 3 (Invertibility)**.** *Suppose B and B' are both inverses of A. Then $B = BI = B(AB') = (BA)B' = IB' = B'$.*

Solution 4 (Invertibility 2)**.** *Suppose the matrix is invertible. Then* $\det \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = 0 \cdot 0 - 1 \cdot 0 = 0$*, but an invertible matrix must have nonzero determinant. Contradiction.*

Solution 5 (Span)**.** $\operatorname{span}(v_1, v_2) = \left\{ \begin{pmatrix} 3x + y \\ x + 3y \end{pmatrix} : x, y \in \mathbb{R} \right\}$.

Solution 6 (Span 2)**.** *Setting up $u = au_1 + bu_2 + cu_3$ gives the augmented matrix* $\left(\begin{array}{ccc|c} 3 & 1 & 2 & 2 \\ 10 & 3 & 8 & 10 \\ 7 & -2 & 1 & 7 \end{array}\right)$*. Gaussian elimination gives $a = \frac{2}{21}$, $b = -\frac{46}{21}$, $c = \frac{41}{21}$, so* $u \in \operatorname{span}(u_1, u_2, u_3)$.

Solution 7 (Subspace criterion)**.** *$U = \{x \in k^n : Ax = 0\}$. Check: $A0 = 0$ so $0 \in U$; if $Ax = Ay = 0$ then $A(x + y) = 0$; if $Ax = 0$ then $A(\lambda x) = 0$. So U is a subspace.*

Solution 8 (Subspace criterion 2)**.** *$U = \{Ax : x \in k^n\}$. Check: $A0 = 0$; $Ax_1 + Ax_2 = A(x_1 + x_2)$; $\lambda Ax = A(\lambda x)$. So U is a subspace.*

Solution 9 (Subspace criterion 3)**.** *If $V \neq \{0\}$, pick $v \in V$, $v \neq 0$. For any $x \in k$, write $x = \lambda v$ for $\lambda = x/v \in k$. By closure under scalar multiplication, $x \in V$. Hence $V = k$.*

Solution 10 (Subspace criterion 4)**.** $0 = 0v_1 + \cdots + 0v_n \in \operatorname{span}$*; sums and scalar multiples of linear combinations are linear combinations. Hence* $\operatorname{span}(v_1, \ldots, v_n)$ *is a subspace.*

Solution 11 (Span membership)**.** *$x = 3$, $y = 1$, $z = -2$, so $v = 3v_1 + v_2 - 2v_3 \in$* $\operatorname{span}(v_1, v_2, v_3)$.

Solution 12 (Span membership 2)**.** *Same system:* $t = 3$, $y = 1$, $z = -2$, *so* $f = 3f_1 + f_2 - 2f_3$.

Solution 13 (Infinite dimension of P_∞)**.** *If* $\dim P_\infty < \infty$, *some* $f_1, \ldots, f_n$ *span* P_∞. *Let* $d = \max \deg(f_i)$. *Then every polynomial in the span has degree* $\leq d$, *but* x^{d+1} *does not. Contradiction.*

Solution 14 (Linear independence)**.** $3v_1 + 5v_2 - 2v_3 = 0$, *so the vectors are not LI.*

Solution 15 (Coordinate vectors)**.** $\lambda_1 = 2$, $\lambda_3 = 0$, $\lambda_2 = -\frac{1}{3}$. *So* $[g]_S = \begin{pmatrix} 2 \\ -1/3 \\ 0 \end{pmatrix}$.

Solution 16 (Coordinate vectors 2)**.** *Evaluate:* $f(1) = -1$, $f(2) = 5$, $f(3) = 15$. *So* $[f]_e = \begin{pmatrix} -1 \\ 5 \\ 15 \end{pmatrix}$.

Solution 17 (Finding a Basis)**.** *Form the matrix with* v_1, v_2, v_3, v_4 *as columns and row reduce:*

$$\begin{pmatrix} 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 8 \\ 1 & 3 & 0 & 2 \\ 1 & 2 & 1 & 10 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & 1 & 2 \\ 0 & -3 & 0 & 4 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 8 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & -3 & 4 \\ 0 & 0 & 0 & 8 \end{pmatrix}.$$

Pivots are in columns 1, 2, 3, 4, so all four vectors are LI and $\{v_1, v_2, v_3, v_4\}$ *is a basis of* V.

Solution 18 (Finding a Basis 2)**.** *Use the isomorphism* $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$ *to get coordinate vectors* $\begin{pmatrix} 1 \\ 2 \\ 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 4 \\ 5 \\ 5 \\ 7 \end{pmatrix}$. *Row reducing the matrix with these as columns gives a free variable for* M_3, *revealing* $M_3 = 2M_1 + M_2$. *So* $\{M_1, M_2\}$ *is a basis.*

Solution 19 (Finding a Basis 3)**.** *(1)* $A = A^*$ *means* $a_{ij} = \overline{a_{ji}}$. *The diagonal entries satisfy* $a_{ii} = \overline{a_{ii}}$, *so they are real. The off-diagonal entries satisfy* $a_{12} = \overline{a_{21}}$. *Writing* $z = a_{12}$ *gives the form* $\begin{pmatrix} x & z \\ \bar{z} & y \end{pmatrix}$.

(2) Writing $z = a + bi$ *with* $a, b \in \mathbb{R}$, *the general Hermitian matrix is*

$$x \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + y \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + a \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}.$$

These four matrices are LI and span U, *so they form a basis. Hence* $\dim_{\mathbb{R}} U = 4$.

Solution 20 (Finding a Basis 4)**.** *Since $f \in U$ is even, it is determined by its values at $x = 0, 1, 2$, so $\dim U = 3$. It suffices to show S is LI (then $|S| = \dim U = 3$ implies it is a basis). Suppose $c_1 g_1 + c_2 g_2 + c_3 g_3 = 0$. Evaluating:*

$$x = 0: \ c_1 = 0, \qquad x = 1: \ c_1 + 0 + c_3 = 0 \implies c_3 = 0, \qquad x = 2: \ c_1 + 0 + 0 = 0.$$

(One checks $g_2(1) = 0$, $g_2(2) = 0$, $g_3(1) = 1$, $g_3(2) = 0$.) Hence $c_1 = c_2 = c_3 = 0$ and S is LI, so S is a basis.

Solution 21 (Basis Extension)**.** *If L spans V, done. Otherwise, since L is LI but not spanning, there exists $v_1 \in V \setminus \operatorname{span}(L)$. Then $L \cup \{v_1\}$ is LI. Repeating: at each step we add one vector to L while maintaining LI. Since V is finite-dimensional, this process terminates in at most $\dim V - |L|$ steps, yielding a basis $B \supseteq L$.*

Solution 22 (Similar matrices have equal characteristic polynomials)**.** $\chi_A(\lambda) = \det(A - \lambda I) = \det(Q^{-1}BQ - \lambda I) = \det(Q^{-1}(B - \lambda I)Q) = \det(Q^{-1}) \det(B - \lambda I) \det(Q) = \chi_B(\lambda)$.

Solution 23 (Characteristic polynomial and invertibility)**.** $\det(A) = 0$ *since A is non-invertible, so $\chi(0) = \det(A - 0 \cdot I) = \det(A) = 0$. Hence 0 is a root of χ.*

12 Useful Links

Proofs: https://www.youtube.com/watch?v=V5tUc-J124s